PASHUPATI

Tuning Guide

What every parameter does, and how to tune a Genisom L1W for a site

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For the technician who sets up and tunes a Pashupati robot. You need an SSH shell on the robot computer and some ROS 2 basics.

Revision history

Version	Date	Changes
1.0	2026-10-16	First version: Pashupati with the Genisom L1W

Operating the robot is described in the Pashupati User Manual. The parameter reference at the end of this guide is generated from the parameter files of this Pashupati version, so it lists every parameter with its default value. Units: m, s, m/s, rad/s, deg; 0-1 means a fraction.

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1 How tuning works

1.1 Where a value comes from

Every Pashupati node starts with the defaults of its package. The site file is applied on top, and live changes on top of that.

Layer	Where	Changed by	After a restart
Package defaults	`src/<package>/config/*.yaml`	a Pashupati update	used
Site file	`/etc/pashupati/site.yaml`	you, for this robot and site	used
Live changes	in the running node	handheld, `ros2 param set`, RViz	lost, except the handheld's inspection and people settings

Never change the package defaults on a robot: the next `git pull` conflicts with your change. Put every change in the site file. The env file `/etc/pashupati/pashupati.env` is not for tuning: it says which robot package runs, which extras start and where NETRA is (chapter 13).

The robot driver (`l1w_driver_node`) is not a Pashupati node. Its settings live in the robot package (section 2.4).

1.2 Change a value for good

1. Open the site file: `sudo nano /etc/pashupati/site.yaml`.
2. Find the node's section, or add one like the example below. Indent with spaces, two per level, never with tabs.
3. Restart the stack: `sudo systemctl restart pashupati`.
4. Check the start of the log for warnings: `journalctl -u pashupati -n 100 | grep -i warn`.

```
path_follower_node:
  ros__parameters:
    target_linear_velocity: 0.35
```

A misspelt **node** name is reported at start: *no node is called "..." (typo?).* A file that is not valid YAML stops the start, so the robot never runs on defaults by accident. A misspelt **parameter** name is silently ignored: check that your value arrived with `ros2 param get`.

1.3 Try a value live

```
pashu                                  # ROS shell of the stack
ros2 param get /path_follower_node target_linear_velocity
ros2 param set /path_follower_node target_linear_velocity 0.3
ros2 param dump /path_follower_node     # every value in use
```

A live change applies at once and is lost at the next restart. When a value works, copy it to the site file. Only some nodes react to live changes:

Node	Reacts live to
`path_follower_node`	everything except `control_rate`, `auto_mode_on_service`, `auto_mode_off_service`, `inspection_services`
`obstacle_avoidance_node`	everything
`anomaly_detector_node`	`enabled`, `model`, `interval_sec`, `categories`, `alarm_categories`, `category_descriptions`
`person_detector_node`	`enabled`, `confidence`, `rate_hz`
all other nodes	nothing: `ros2 param set` is accepted, but the node only reads its parameters at start. Use the site file and restart

The **Tuning** panel of RViz (`robot_rviz`) edits the navigation and avoidance parameters with sliders; its **Save YAML...** button writes them to a file. RViz runs on a laptop with ROS 2 Humble on the robot's network: set `ROS_LOCALHOST_ONLY=0` in the env file, restart, and use the robot's `ROS_DOMAIN_ID` and `RMW_IMPLEMENTATION` on the laptop.

> **SET IT BACK**
>
> With `ROS_LOCALHOST_ONLY=0` anyone on the robot's network can send motion commands to the robot. Set it back to `1` and restart when you are done.

1.4 What the handheld changes

On the handheld	Parameter	After a restart
Patrol speed slider	`path_follower_node` `target_linear_velocity`, from `patrol_speed_min` up to `max_linear_velocity`	the site file value
Inspect the floor	`anomaly_detector_node` `enabled`	kept
Pace: Continuous, Interval, Stop points	`interval_sec`: 0, the interval, -1	kept
Model	`model`	kept
Finding types	`categories`, `alarm_categories`, `category_descriptions`	kept
Count people	`person_detector_node` `enabled`	kept
Speed 1, 2, 3	30, 60, 100 % of `teleop_udp_node` `max_vx`, `max_vy`, `max_wz`	a handheld setting
Inspection at a stop point	the dwell time stored in the route with each **Point**	in the route

Kept values are stored in `~/.pashupati/inspection.json` and `~/.pashupati/people.json` and **win over the site file**. To go back to the site file's inspection settings, stop the service, delete `inspection.json`, and start it again.

1.5 Tuning a new site, step by step

1. **The robot, once:** body size, body height, speed limits, stick channels (chapter 2).
2. **Braking:** measure it and test the protective field (chapter 3).
3. **AprilTags:** print, place, configure and record them (chapter 8).
4. **First route:** teach a short route and patrol it at 0.3 m/s (chapters 4 to 6).
5. **Obstacles:** a cardboard box on the route, then a person walking past (chapter 7).
6. **Full speed:** raise the patrol speed step by step, one full lap per step.
7. **AI and people:** pace, model, finding types, people counting (chapters 10 and 11).

8. **Save:** copy every value that works into the site file, restart, drive one more lap.

1.6 Work safely

- Tune in a closed area first, never in a crowd.
- Keep the ZSBot remote in reach, and someone on the handheld's STOP.
- Change one thing at a time. Write down the old and the new value.
- Start low: speeds and accelerations below what you need, then raise them step by step.
- Repeat the protective field test (section 3.4) after every change to the body size, braking or speed limits.

2 The Genisom L1W

2.1 Body size

Pashupati treats the robot as a box around `base_link`, the middle of the body. Three nodes use that box, and their values must agree:

Node	Parameters	Used for
`control_mux_node`	`field.front`, `field.rear`, `field.half_width`	protective field (chapter 3)
`path_follower_node`	`footprint.front`, `footprint.rear`, `footprint.half_width`	path shift around obstacles (section 7.1)
`lidar_sector_node`	`self_filter_x_min`, `self_filter_x_max`, `self_filter_y_min`, `self_filter_y_max`	the robot's own legs, ignored by the LiDAR

On the L1W the box is 0.80 × 0.56 m: front 0.40, rear 0.40, half width 0.28; self filter x from -0.40 to 0.40 and y from -0.28 to 0.28. Measure again when the robot carries a payload or a cover that sticks out.

2.2 Body height

`base_link` is about 0.35 m above the floor when the L1W stands. Two parameters follow from it:

Parameter	L1W	Rule
`lidar_sector_node` `obstacle_z_min`	-0.25	minus the height of `base_link`, plus 0.10: anything less than 10 cm tall counts as floor
`person_detector_node` `floor_z`	-0.35	minus the height of `base_link`: where people's feet are placed on the map

To measure the height of `base_link`, stand the robot on a flat floor, measure the height of the LiDAR above the floor, and subtract the LiDAR's height in the body: `ros2 run tf2_ros tf2_echo base_link livox_frame` (the z value).

2.3 Speed limits

Parameter	L1W	Limits
`control_mux_node` `max_vx`, `max_vy`, `max_wz`	1.0, 0.5, 1.5	every command, AUTO and REMOTE: nothing may command more
`teleop_udp_node` `max_vx`, `max_vy`, `max_wz`	1.0, 0.5, 1.5	the handheld at speed level 3 (100 %)
`path_follower_node` `max_linear_velocity`	0.6	the patrol, and the top of the handheld's patrol speed slider
`l1w_driver_node` `stick_max_vx`, `stick_max_vy`, `stick_max_wz`	1.0, 0.5, 1.5	the command that means a full stick (next section)

`max_vy` is 0 for a robot that cannot walk sideways. The L1W can, so keep `holonomic: true` in `path_follower_node`.

2.4 Stick channels and speed scaling

The L1W takes motion as joystick values, like its own remote. `l1w_driver_node` turns each speed command into a stick position: a command of `stick_max_vx` is a full stick forward. Its settings are in the robot package, `src/xtras/robots/genisom_l1w_ros2/l1w_driver/config/l1w_driver.yaml`. Edit them there and restart the service; they are not live.

Parameter	L1W	Meaning
`stick_max_vx`, `stick_max_vy`, `stick_max_wz`	1.0, 0.5, 1.5	speed that maps to full stick deflection
`stick_yaw_axis`, `stick_lateral_axis`	2, 1	stick channels for turning and walking sideways

Parameter	L1W	Meaning
`stick_invert_yaw`, `stick_invert_lateral`	false	true when the robot turns or walks sideways the wrong way
`cmd_timeout`	0.5	s; without a new command the stick returns to zero
`auto_control_right`	true	take motion control from the ZSBot remote when the driver connects

Check the directions once (REMOTE, speed level 1): left stick to the left must walk the robot to its left; right stick to the left must turn it to the left. If one is wrong, set its `stick_invert_*` to true.

Check the speed scale: the speed the L1W reaches at full stick depends on its speed gear (*Slow*, *Normal*, *Fast speed* in the robot actions). Patrol a straight route and compare the patrol speed with the measured speed: `ros2 topic echo /l1w/odom --field twist.twist.linear.x`. If the robot is slower or faster, set `stick_max_vx` to its value × measured speed ÷ patrol speed, and scale `stick_max_vy` and `stick_max_wz` the same way. Always patrol in the gear you checked.

> **NOTE**
>
> The change is local to the robot package's clone. Commit it to `genisom_l1w_ros2` when it holds for every L1W, or note it in the robot's logbook so an update does not undo it.

3 Protective field

3.1 How it works

The control mux limits every motion command to a speed at which the robot can still stop `field.margin` before the nearest obstacle in the path it is about to sweep. It looks in every direction (360°, from the LiDAR's safety scan):

$$v \times t + v^2 \div 2a \le \textit{free distance} - \textit{margin}$$

v speed · t `field.reaction_time` · a `field.max_decel` · *margin* `field.margin`

The free distance is measured from the robot's edge (chapter 2). With the defaults the robot may move at most:

Free distance from the edge	0.3 m	0.5 m	0.75 m	1.0 m	1.5 m	2.0 m
Top speed (m/s)	0.30	0.52	0.71	0.87	1.13	1.35

The field works in AUTO. In REMOTE the operator watches the robot and nothing limits it, unless `field_in_remote` is true. Without a fresh scan the robot does not drive itself at all.

3.2 Parameters

All in `control_mux_node`, read at start:

Parameter	Default	What it does
`field_in_auto`	true	limit AUTO (the patrol). Keep it on
`field_in_remote`	false	limit REMOTE too, e.g. for operators new to the robot
`field.max_decel`	0.6	m/s², braking the robot reliably achieves: **measure it** (section 3.3)
`field.reaction_time`	0.25	s, from the command to the robot actually braking
`field.margin`	0.15	m, gap left to the obstacle
`field.lateral_margin`	0.10	m, extra width of the swept path on each side

Parameter	Default	What it does
`field.min_speed`	0.02	m/s, slower commands are not limited (creeping, turning)
`safety_scan_timeout_sec`	0.5	s; an older scan stops AUTO and caps REMOTE
`stale_remote_speed`	0.15	m/s, the REMOTE cap while the scan is stale

A lower `max_decel`, a longer `reaction_time` or a bigger `margin` make the robot slow down earlier and stop further from obstacles. Too low, and it crawls in narrow aisles.

3.3 Measure the braking

1. Teach a straight route of at least 8 m, with free space beyond it.
2. Set the patrol speed on the handheld to the fastest you will patrol at, for example 0.6.
3. Put a tape mark on the floor in the middle of the route.
4. Start the patrol. Press **STOP** when the front of the robot passes the mark.
5. Measure the distance *d* from the mark to the front of the stopped robot.
6. max_decel = speed^2 ÷ (2 × (*d* − speed × 0.25)).
7. Repeat three times. Use the smallest result.

Your own reaction is part of *d*, so the result errs on the safe side. At 0.6 m/s:

Stopping distance *d*	0.40 m	0.50 m	0.60 m	0.80 m	1.00 m
`field.max_decel`	0.72	0.51	0.40	0.28	0.21

Measure on every new kind of floor: a polished floor brakes worse than a rough one.

3.4 Test it

1. In the site file set `field_in_remote: true` and restart.
2. Put a cardboard box in an open area.
3. In REMOTE at speed level 1, drive straight at the box. The robot must slow down and stop about `field.margin` (15 cm) before it. Try level 2, then from the side and backwards.
4. Set `field_in_remote` back to false (or keep it on) and restart.

If the robot gets closer than the margin, lower `field.max_decel` or raise `field.reaction_time`. Never test with a person in front of the robot.

3.5 Control mux

Parameter	Default	What it does
`remote_timeout_sec`	0.3	s; when the handheld goes silent the robot gets zero speed
`nav_timeout_sec`	0.5	s; the same for navigation in AUTO
`pause_nav_on_remote`	true	switching to REMOTE pauses the patrol
`estop_hold_hz`	10	zero commands per second while STOP is engaged

4 Speed and motion

4.1 Cruise speed and curves

All in `path_follower_node`, live:

Parameter	Default	What it does · when to change
`target_linear_velocity`	0.4	m/s, cruise speed; the handheld's patrol speed slider sets it. 0.3 in crowds
`max_linear_velocity`	0.6	m/s, highest cruise speed, also for the slider
`min_linear_velocity`	0.1	m/s, the speed profile never goes lower
`max_lateral_accel`	0.3	m/s^2, speed in curves: lower = slower in curves, steadier
`decel_limit`	0.3	m/s^2, braking planned into stop points and the route end
`approach_speed`	0.15	m/s, speed when arriving at a stop point or the route end
`max_angular_velocity`	1.0	rad/s, turn rate cap

The speed in a curve is $\sqrt{}$(`max_lateral_accel` × radius): with 0.3, a curve of 1 m radius is walked at 0.55 m/s and one of 0.5 m radius at 0.39 m/s.

4.2 Acceleration and smoothness

Parameter	Default	What it does · when to change
`linear_accel_limit`	0.5	m/s^2, how fast the speed may change. Jerky starts: lower it. A hard stop ignores it
`angular_accel_limit`	1.5	rad/s^2, the same for turning
`command_smoothing`	0.4	filter on turning and sideways commands: 1 = off, lower = smoother but slower to react. The body wags: lower it
`control_rate`	20	Hz, control loop rate (read at start)

5 Following the route

5.1 Holonomic controller

`controller: holonomic` is the controller for legged robots. The L1W walks forward, sideways or backwards as the route needs and keeps its heading along the route.

Parameter	Default	What it does · when to change
`holonomic.lookahead_distance`	0.6	m, point on the route ahead the robot walks to. Longer: smoother, gentler return to the route, cuts corners more. Shorter: follows tightly, may weave
`holonomic.heading_lookahead`	0.6	m, the heading follows the route direction this far ahead. Longer: turns earlier into curves
`holonomic.kp_yaw`	1.5	heading correction strength. The body wags: lower it. Heading lags in curves: raise it
`holonomic.max_lateral_velocity`	0.25	m/s, sideways speed cap
`holonomic.max_reverse_velocity`	0.15	m/s, backwards speed when the route is behind the robot
`holonomic.min_alignment_scale`	0.25	speed factor while the heading is far off: low = turns first, then walks
`holonomic.goal_tolerance`	0.3	m, the end of an open route counts as reached

5.2 Leaving and joining the route

Parameter	Default	What it does
`off_path_slow_distance`	0.6	m from the route: slow down to `min_linear_velocity`

Parameter	Default	What it does
`off_path_stop_distance`	1.5	m from the route: stop and report *Off the route*; the patrol needs a new **Start**
`join_max_distance`	3.0	m; started up to this far from the route, the robot first walks onto it. 0 = off
`join_timeout_sec`	60	s, give up walking onto the route and follow from where the robot is

Sliding around an obstacle (section 7.1) does not count as being off the route.

5.3 Route smoothing

`path_loader_node` smooths a recorded route between its stop points when it loads it. Stop points never move.

Parameter	Default	What it does · when to change
`smoothing_enabled`	true	smooth the route
`resample_spacing`	0.1	m between route points after smoothing
`smooth_weight_data`	0.05	pull toward the recording: higher keeps more of the wiggles you drove
`smooth_weight_smooth`	0.4	pull toward the neighbours: higher is smoother. Keep it below 0.5
`max_smoothing_deviation`	0.3	m; the log warns when smoothing moved the route further

If the smoothed route cuts a corner too close to a wall, lower `smooth_weight_smooth`, or record that part again more carefully.

5.4 Other controllers

`pure_pursuit`, `pid`, `mppi`, `lqr` and `stanley` only steer (forward and turn), for robots that cannot walk sideways. With them, `speed_regulator_enabled` slows down in sharp turns. On the L1W keep `holonomic`. Their gains are listed in the parameter reference.

6 Stop points

6.1 What happens at a stop

1. The robot slows down to `approach_speed` before the stop point (`decel_limit`).
2. Within `approach_radius` it switches to the final approach: it moves straight onto the recorded position and heading, walking sideways when needed.
3. It calls the inspection services and starts the dwell time stored in the route.
4. It moves on when the dwell time is over and the inspection answered, or after the dwell time plus `inspection_timeout_sec` .

A stop point the robot has already passed by more than `stop_point_skip_margin` route points (0.5 m at the default spacing) is skipped, for example after a long detour.

6.2 Final approach

Parameter	Default	What it does · when to change
`approach_radius`	0.5	m from the stop point where the final approach starts
`approach.kp_xy`	1.2	position correction strength
`approach.kp_yaw`	1.5	heading correction strength
`approach.max_speed`	0.2	m/s
`approach.max_yaw_rate`	0.6	rad/s
`approach.min_speed`	0.04	m/s, lowest command. Must beat the joystick deadband: when the robot stops just short of the point, raise it to 0.05-0.08
`approach.position_tolerance`	0.05	m, close enough
`approach.yaw_tolerance_deg`	3	deg, close enough
`approach_timeout_sec`	20	s; then the robot gives up and inspects where it is (the log says *approach timeout*)

6.3 Inspection at stops

Parameter	Default	What it does
`inspection_services`	`[/anomaly_detector_node/check_now]`	services called at every stop point (read at start)
`inspection_timeout_sec`	20	s the robot waits for them beyond the dwell time

The dwell time comes from the route: the handheld's *Inspection at a stop point* setting is stored with each **Point** you mark (10 s unless changed).

7 Obstacles

7.1 Path shift

With `avoidance_mode: path_shift` (the default with the holonomic controller) the robot passes an obstacle by sliding sideways along a line parallel to the route, heading kept. When there is no gap wide enough on either side, it slows down and waits in front of the obstacle. After `blocked_timeout_sec` it reports *Blocked* and keeps waiting until the way is free.

Parameter	Default	What it does · when to change
`path_shift.max_offset`	0.8	m, furthest the robot leaves the route. Narrow aisles: lower it
`path_shift.look_ahead`	2.0	m of route ahead of the front edge that must be free
`path_shift.clearance`	0.10	m kept between the robot's side and an obstacle. It does not pass gaps that look wide enough: lower it a little
`path_shift.offset_rate`	0.3	m/s, how fast it slides sideways
`path_shift.switch_cost`	0.5	higher: keeps its side longer instead of switching
`path_shift.switch_margin`	0.05	m, extra room a new side needs (no flicker)
`path_shift.return_delay_sec`	0.5	s, the way back to the route must stay free this long
`path_shift.decel`	0.5	m/s^2, braking toward an obstacle it cannot pass
`path_shift.stop_margin`	0.40	m, stops this far before an obstacle (from the front edge)
`blocked_timeout_sec`	30	s without progress before *Blocked* is reported

7.2 Hard stop

`obstacle_avoidance_node` stops the robot at once, in every mode, when anything is closer than `emergency_distance` to `base_link` inside the front cone of `emergency_cone_deg`. The robot continues when the way stayed free for `estop_release_sec` (1 s, `path_follower_node`).

Parameter	Default	What it does
`emergency_distance`	0.75	m from `base_link`: about 0.35 m in front of the L1W's nose
`emergency_cone_deg`	60	deg, width of the front cone
`safe_distance`	1.0	m; closer obstacles shape the command (reactive modes only)

Keep `footprint.front` + `path_shift.stop_margin` (0.80 m) above `emergency_distance` (0.75 m): the robot then normally stops smoothly in front of an obstacle, and the hard stop only catches what appears suddenly.

7.3 What the LiDAR sees

`lidar_sector_node` turns the LiDAR points into the obstacle scans. Read at start.

Parameter	Default	What it does · when to change
`input_type`	livox	`livox` reads the Livox driver directly; keep it on the L1W
`obstacle_z_min`	-0.25	m in `base_link`: lower points are floor. Floor or ramps seen as obstacles: raise it. Low objects missed: lower it
`obstacle_z_max`	1.20	m in `base_link`: higher points are ignored (signs, ceilings)
`self_filter_*`	±0.40, ±0.28	m, the robot's own body. *Slowed* with nothing around: the legs are seen, widen it a little
`range_min`, `range_max`	0.2, 8.0	m, LiDAR range used

Parameter	Default	What it does · when to change
`safety_num_sectors`	72	sectors of the 360° safety scan (5° each)
`safety_range_max`	4.0	m; further obstacles never limit the speed
`min_raw_points`	100	fewer points per scan means the LiDAR is covered or failed: everything counts as blocked

> **WARNING**
> The LiDAR does not see drops (stairs, platform edges), glass, mirrors or thin poles well. Keep routes away from them; no parameter fixes that.

7.4 Reactive avoidance

`avoidance_mode: reactive` (and every non-holonomic controller) uses the steering of `obstacle_avoidance_node` instead of the path shift. `algorithm` picks it: `vfh` (default), `braitenberg`, `potential_field` or `follow_gap`. Each has a `steering_gain` (how hard obstacles turn the robot), a `velocity_gain` (how much they slow it) and a `smoothing` filter (0-1, higher is smoother and slower). The L1W uses the path shift; the gains are in the parameter reference.

8 Localization

8.1 AprilTags

The robot finds itself on the map from AprilTags: printed squares whose positions on the map are saved in `tag_config.json` (`TAGS` in the env file). The `apriltag` node in `perception_params.yaml` must know every tag:

Parameter	Default	What it does
`family`	36h11	tag family; print tags of this family
`tag.ids`	[2, 4]	IDs of the tags in use
`tag.frames`	[base_map, charging_dock]	a name for each ID, in the same order
`tag.sizes`	[0.15, 0.15]	m, edge of each tag's black square, in the same order
`size`	0.16	m, edge of any other tag
`detector.decimate`	2.0	image shrink before detection: 1.0 sees tags further away, uses more CPU
`max_hamming`	0	bit errors allowed; keep 0 (no false tags)
`apriltag_throttle rate_hz`	5	camera frames per second given to the detector

The three `tag.*` lists must have the same length, and `localization_node` `tag_frames` must list the same names.

8.2 Placing and recording tags

- Measure the printed size across the black square, not the white margin, and enter it in `tag.sizes`. A wrong size puts the robot in the wrong place.
- Mount tags flat and rigid on walls at about the camera's height, where the robot stands still facing them: at the start of the route, and at places where it should correct drift.
- Bigger tags are seen from further away. Avoid glare and direct sunlight on the tag.

- After mounting, record them from the handheld (**Tags**, see the User Manual). The positions are averaged over `tag_record_duration_sec` and saved to `tag_config.json`; the old file is kept as a `.bak`. Moving a tag later means recording it again.

8.3 Localization node

Parameter	Default	What it does · when to change
`auto_localize_on_start`	true	try to localize from a tag every second after start, until it works
`assume_start_origin`	false	true: no tag needed, the start pose is the map origin. Then the robot must always start at exactly the same spot and heading
`auto_correct_period_sec`	0	s; > 0 re-corrects drift from any visible tag this often. 30 for long patrols past tags
`max_correction_m`	0.5	m; larger automatic corrections are rejected
`max_correction_deg`	15	deg, the same for heading
`tag_max_age_sec`	0.5	s; older tag sightings are ignored
`map_odom_hz`	20	how often the map correction is republished

8.4 Odometry checks

FAST-LIO, the LiDAR odometry, tracks the robot between tags. It needs a still, level start: `odometry_gate` starts it only when the LiDAR has been still for `still_sec` (2 s) and tilts less than `max_tilt_deg` (8°). After `timeout_sec` (60 s) it starts anyway and the health status reports an error. Give the robot a still moment after standing up.

`odometry_monitor_node` watches the running odometry and reports to `/health/status`:

Parameter	Default	Reports when
`stale_sec`	0.5	no odometry for this long
`max_speed_mps`	3.0	the pose jumps faster than this

Parameter	Default	Reports when
`max_z_drift_m`	0.5	the height drifts (tilted start). 0 = off
`max_speed_diff_mps`	0.35	LiDAR and leg speeds differ this much …
`speed_diff_sec`	1.5	… for this long (slipping, featureless corridor)
`restart_origin_radius_m`, `restart_min_offset_m`	0.05, 0.3	FAST-LIO restarted: the pose jumped back to its origin

These rarely need tuning. A *speed mismatch* in one long, plain corridor is a hint to add a tag there and use `auto_correct_period_sec`.

9 Patrol loop and fail-safes

All in `path_follower_node`, live:

Parameter	Default	What it does
`loop_route`	true	restart a closed route when it ends
`loop_close_distance`	1.0	m; a route whose end is this close to its start is closed
`low_battery_percentage`	0.2	0-1; below it the patrol stops at the end of the lap
`require_localization`	true	refuse **Start** until the robot is localized
`estop_release_sec`	1.0	s the way must stay clear after a hard stop
`avoidance_timeout_sec`	0.5	s; stop when the obstacle scan goes silent
`tf_timeout_sec`	0.5	s; stop when the robot pose stops updating (odometry died)
`auto_mode_on_service`, `auto_mode_off_service`	""	services called when a patrol starts and stops (read at start)

On the L1W `auto_mode_*_service` stay empty: the driver takes control when it connects (`auto_control_right`), and **Take control** on the handheld takes it back after the ZSBot remote was used.

10 AI inspection

10.1 Pace and queue

`anomaly_detector_node` sends camera images to an AI vision model (Ollama, on the robot) and gets back one finding type or *none*.

Parameter	Default	What it does · when to change
`enabled`	true	live; off also unloads the model from memory
`interval_sec`	0	live: 0 continuous, > 0 one image every N s, < 0 only at stop points
`queue_max`	50	interval pace: images waiting for the model; when full, the oldest is dropped
`queue_min_change`	12	grey levels the picture must change to be queued again: standing still adds nothing. 0 = every image
`queue_dir`	`~/.pashupati/inspection_queue`	waiting images survive a restart; "" = memory only
`cooldown_sec`	60	s; continuous and interval pace do not report the same type again within this time
`max_image_age_sec`	2.0	s; older camera images are not inspected
`max_image_width`	640	px; images are shrunk first. Smaller is faster but misses small things

One inspection takes about 1-2 s on the Jetson. An interval shorter than that fills the queue; that is fine while the robot stands at stops, where the queue drains.

10.2 Model and memory

Parameter	Default	What it does
`model`	qwen3-vl:2b-instruct	live; the model in use
`models`	three models	offered on the handheld (plus any other vision model in Ollama)
`keep_alive`	"-1"	keep the model loaded: "-1" forever, or e.g. "30m" (quoted)
`num_ctx`	2048	tokens of context: the image and the type list must fit. Many long descriptions: 4096 (more memory)
`timeout_sec`	30	s, longest wait for one answer
`use_schema`	true	structured answers; false only for models that cannot do them
`describe`	false	true: the model also writes one sentence about the floor (slower)
`debug`	false	true: log every inspection with the raw model answer

A model must be pulled before it can be used (`ollama pull <model>`, or `script/setup.sh install ollama` with `VLM_EXTRA_MODELS`). `setup.sh install ollama` also turns off Ollama's prompt cache, which otherwise grows to several GB of RAM. Check with `free -h`: with the model loaded the robot normally uses about 7 GB of its 15 GB.

10.3 Finding types

Parameter	Default	What it does
`categories`	[trash, spill, stain]	live; the types the model chooses from (plus *none*), up to 12, a-z 0-9 _
`alarm_categories`	[trash, spill]	live; only these are reported to NETRA, the rest is shown on the handheld
`category_descriptions`	[""]	live; own text per type, `"name: what to look for"`

`trash`, `spill`, `stain`, `floor_damage` and `fallen_person` have built-in descriptions; any other type needs one. The handheld edits all three lists (**Inspect → Finding types**); its changes win over the site file (section 1.4). Good descriptions:

- are in English, short and concrete: *a bag, backpack or suitcase standing alone on the floor*;
- name what is visible in the image, not what it means;
- say what is **not** a finding when the model confuses things: *not shadows, reflections or floor patterns*.

Too many false *stain* findings on a shiny floor is why `stain` is not reported by default.

10.4 Reporting

`fleet_uplink_node` sends findings to NETRA:

Parameter	Default	What it does
`anomaly_min_confidence`	0.5	0-1; less confident findings are not sent
`spool_dir`	`~/.pashupati/spool`	findings wait here while NETRA cannot be reached

11 People counting

`person_detector_node` finds people in the camera image with YOLO11n. Started only with `PEOPLE=true` in the env file; the handheld switches it on and off.

Parameter	Default	What it does · when to change
`enabled`	true	live; on or off
`confidence`	0.4	live, 0-1. Boxes on things that are not people: raise it. People missed: lower it
`rate_hz`	0	live; detections per second. 0 = 5 on the GPU, 2 on the CPU
`threads`	2	CPU threads for one detection (CPU only)
`floor_z`	-0.35	m, floor height in `base_link` (section 2.2)
`max_range`	12	m; people further away are not placed on the map
`backend`	auto	`onnxruntime`, `opencv` or `ultralytics`; auto picks the first that works
`model_path`	`models/yolo11n-416.onnx`	installed by `script/setup.sh install people_model`

NETRA's crowd map uses `fleet_uplink_node` `crowd_cell_size` (1 m grid) and `crowd_window_sec` (one sample per minute, the peak per cell).

12 Handheld, video, NETRA and health

12.1 Handheld link

`teleop_udp_node`, read at start:

Parameter	Default	What it does
`token`	""	shared secret; the same in the handheld's connection settings. **Always set it**
`robot_name`	""	name shown on the handheld's start screen (empty: the computer's name)
`port`	9870	UDP port of the link and of robot discovery
`client_timeout_sec`	2.0	s; a silent handheld is dropped
`robot_commands`	see chapter 15	the robot actions on the handheld, in this order
`patrol_speed_min`	0.1	m/s, slowest patrol speed the slider offers
`lidar_hz`, `lidar_voxel`, `lidar_max_points`	2, 0.1, 3000	LiDAR points on the handheld's map: fewer is lighter on Wi-Fi

12.2 Video

The handheld gets H.264 video over WebRTC from `video_stream_node`, started while it watches:

Parameter	Default	What it does
`bitrate_kbps`	1500	weak Wi-Fi, stuttering video: 800
`fps`	15	frames per second: 10 saves Wi-Fi and CPU
`max_width`	1280	px; wider images are shrunk
`keyframe_sec`	1.0	s; a new viewer starts at a key frame, keep it short
`encoder`	auto	the Jetson's hardware encoder when it works

Without WebRTC (`WEBRTC=false` or MediaMTX missing) the handheld gets JPEG frames instead: `teleop_udp_node` `camera_fps` (8), `camera_jpeg_quality` (60), `camera_max_width` (640). The camera itself runs at 640 × 480, 15 fps (`l1w_bringup/config/realsense_params.yaml`).

12.3 NETRA uplink

`fleet_uplink_node` :

Parameter	Default	What it does
`server_url`	""	NETRA address; set by `CC_URL` in the env file
`token`	""	the robot's token from NETRA; set by `CC_TOKEN`
`robot_id`	robot-01	the robot's name in NETRA; `ROBOT_ID` in the env file wins
`telemetry_hz`	1.0	position and status messages per second
`upload_period_sec`	5.0	s between upload attempts of waiting findings and routes
`verify_tls`	true	check NETRA's certificate; keep it on

12.4 Health monitor

`health_monitor_node` publishes `/health/status` , shown on the handheld and in NETRA:

Parameter	Default	Issue when
`topics`	LiDAR, odometry, scans, camera, robot odometry	a topic runs below 80 % of its listed rate (`"topic:Hz"`)
`max_cpu_temp_c`	85	the CPU is hotter
`max_memory_percent`	92	more memory is in use
`min_disk_free_gb`	5	less disk is free
`startup_grace_sec`	30	s after start without missing-node issues

13 The env file

`/etc/pashupati/pashupati.env` says what runs. Restart the service after a change. Quote values with spaces, and never put a comment after a value.

Key	Genisom L1W	Meaning
`ROBOT_ID`	e.g. `l1w-01`	the robot's name in NETRA
`ROBOT_BRINGUP`	`l1w_bringup`	robot package started with the stack
`ROBOT_NS`	`l1w`	namespace of the driver's topics and services
`ROBOT_PRESTART`	`"/usr/local/bin/robot-launch stop driver-livox forward"`	stops the vendor's own LiDAR driver and motion stack first
`ROBOT_SETUP`	`…/genisom_l1w_ros2/script/setup.sh`	the robot package's installer (ZSBot SDK)
`ROBOT_IP`	`192.168.234.1`	motion computer, for the network check
`MODE`	`patrol`	`teach` : no navigation, the LiDAR map is saved
`ODOMETRY`	`fast_lio`	or `leg_kilo`
`ROUTE`	empty	route loaded at start; empty: the last one loaded
`TAGS`	`…/map/tag_config.json`	AprilTag positions
`ROS_DOMAIN_ID` , `RMW_IMPLEMENTATION`	`0` , `rmw_fastrtps_cpp`	Pashupati's own DDS; the vendor stack uses another

Key	Genisom L1W	Meaning
`ROS_LOCALHOST_ONLY`	`1`	`0` only for RViz from a laptop (section 1.3)
`PEOPLE`, `INSPECTION`	`true`	start people counting and the AI inspection at all
`SITE_CONFIG`	`/etc/pashupati/site.yaml`	the site file
`WEBRTC`, `MEDIAMTX`	`true`, empty	WebRTC video for the handheld
`CC_URL`, `CC_TOKEN`	NETRA's address, the robot's token	NETRA uplink; empty: findings are kept on the robot
`TIME_SYNC_WAIT`	`60`	s to wait at boot for the network clock
`OPENBLAS_NUM_THREADS`, `OMP_NUM_THREADS`	`1`	keep at 1: more threads waste CPU

`script/setup.sh check` reports a missing key, or a line that breaks the rules above.

14 Symptoms and fixes

Symptom	Look at	Try
Weaves left and right on straight parts	`holonomic.lookahead_distance`, `holonomic.kp_yaw`, `command_smoothing`	lookahead 0.8, kp_yaw 1.0, smoothing 0.3
Swings wide or cuts corners	`holonomic.lookahead_distance`, `max_lateral_accel`	lookahead 0.4-0.5, lateral accel 0.2
Too slow in gentle curves	`max_lateral_accel`	0.4
Jerky starts and stops	`linear_accel_limit`, `decel_limit`	0.3, 0.2
Walks slower or faster than the patrol speed	`stick_max_*` (section 2.4), speed gear	rescale; patrol in the checked gear
Turns or walks sideways the wrong way	`stick_invert_yaw`, `stick_invert_lateral`	true
Stops just short of a stop point and waits 20 s	`approach.min_speed`, `approach.position_tolerance`	0.06, 0.08
Leaves stop points too late	dwell time of the route, `inspection_timeout_sec`, AI pace	shorter dwell, faster model
Slowed or *Blocked* with nothing around	`self_filter_*`, `obstacle_z_min`, floor slopes	widen the filter 2-3 cm; raise z_min 0.03
Ignores low objects (bags, boxes)	`obstacle_z_min`	lower 0.03 at a time, check the floor is not seen

Symptom	Look at	Try
Stops far before obstacles	`path_shift.stop_margin`, `field.margin`, `field.max_decel`	small steps; repeat the test of section 3.4
Gets too close to obstacles	`field.max_decel`, `field.reaction_time`	measure the braking again (section 3.3)
Does not pass through gaps that look wide enough	`path_shift.clearance`, `path_shift.max_offset`, `footprint.half_width`	clearance 0.05, larger offset
Switches sides around an obstacle	`path_shift.switch_cost`, `path_shift.switch_margin`	1.0, 0.10
Blocked too soon or too late	`blocked_timeout_sec`	to taste
Off the route after a push	`off_path_stop_distance`	2.0, if the area allows
Start refused: robot too far from the route	`join_max_distance`	drive closer, or raise it
No loc: tag not found	`tag.sizes`, `tag.ids`, light, `detector.decimate`	measure the tag, decimate 1.0
Position drifts over long patrols	`auto_correct_period_sec`	30, with tags along the route
Same trash reported again and again	`cooldown_sec`	120-300
False findings	type description, `alarm_categories`, `anomaly_min_confidence`	describe what is not a finding; 0.6-0.7

Symptom	Look at	Try
Inspection queue always full	`interval_sec`, `model`, `max_image_width`	longer interval, smaller model
RAM near the limit	Ollama prompt cache, `keep_alive`, extra models	`setup.sh install ollama`; inspection off unloads the model
People count jumps	`confidence`, `rate_hz`	0.5, 2
Video stutters	`bitrate_kbps`, `fps`	800, 10
Handheld map lags	`lidar_hz`, `lidar_voxel`, `lidar_max_points`	1, 0.2, 1500
Findings missing in NETRA	`CC_URL`, `CC_TOKEN`, `alarm_categories`, `anomaly_min_confidence`	check the log of `fleet_uplink_node`

15 Site file for the Genisom L1W

A complete starting point for `/etc/pashupati/site.yaml`, in two parts: the robot, and the site. Measure and change the values marked MEASURE. Both parts go in the same file.

15.1 The robot part

```
control_mux_node:
  ros__parameters:
    field.front: 0.40
    field.rear: 0.40
    field.half_width: 0.28
    field.max_decel: 0.6          # MEASURE on this floor (3.3)
    field.reaction_time: 0.25
    max_vx: 1.0
    max_vy: 0.5
    max_wz: 1.5

path_follower_node:
  ros__parameters:
    holonomic: true
    target_linear_velocity: 0.35  # cruise speed for this site
    low_battery_percentage: 0.25

lidar_sector_node:
  ros__parameters:
    obstacle_z_min: -0.25         # MEASURE: -(base_link height) + 0.10
    self_filter_x_min: -0.40
    self_filter_x_max: 0.40
    self_filter_y_min: -0.28
    self_filter_y_max: 0.28

person_detector_node:
  ros__parameters:
    floor_z: -0.35                # MEASURE: -(base_link height)

teleop_udp_node:
  ros__parameters:
    token: "change-me"            # the same in the handheld
    max_vx: 1.0
    max_vy: 0.5
    max_wz: 1.5
    robot_commands: [stand_up, sit_down, balance_stand_mode,
                     move_mode, lock_mode, slow_speed,
                     normal_speed, fast_speed,
                     sdk_control_right, emergency_stop]
```

15.2 The site part

```
obstacle_avoidance_node:
  ros__parameters:
    emergency_distance: 0.75

apriltag:                          # the tags on this site's walls
  ros__parameters:
    tag:
      ids: [2, 4]
      frames: [base_map, charging_dock]
      sizes: [0.15, 0.15]          # MEASURE the printed square

localization_node:
  ros__parameters:
    tag_frames: [base_map, charging_dock]   # as tag.frames
    auto_correct_period_sec: 0.0

fleet_uplink_node:
  ros__parameters:
    robot_id: l1w-01               # ROBOT_ID in the env file wins
```

Each node may appear only once in the file: add more parameters under its existing heading. After a restart, `ros2 param dump /control_mux_node` (and so on) shows the values in use.

16 Parameter reference

Every parameter of every Pashupati node with its default value, read from the parameter files of this version. *Meaning* is the comment in the file. Live parameters are listed in section 1.3; change the others in the site file and restart.

16.1 robot_navigation

route following, avoidance, stops · `src/robot_navigation/config/navigation_params.yaml`

path_follower_node

Parameter	Default	Meaning
`control_rate`	`20.0`	Hz, control loop frequency
`target_linear_velocity`	`0.4`	m/s, cruise speed (keep low in crowds)
`max_linear_velocity`	`0.6`	m/s, hard cap
`min_linear_velocity`	`0.1`	m/s, floor of the speed profile (never crawl slower)
`max_lateral_accel`	`0.3`	m/s^2, curvature speed limit v = sqrt(a / kappa)
`decel_limit`	`0.3`	m/s^2, braking used to plan speed into stops and the end
`approach_speed`	`0.15`	m/s, speed when arriving at stop points and the route end
`max_angular_velocity`	`1.0`	rad/s, hard cap
`linear_accel_limit`	`0.5`	m/s^2, ramp limit (emergency stops bypass it)
`angular_accel_limit`	`1.5`	rad/s^2, ramp limit
`command_smoothing`	`0.4`	0-1 low-pass on angular/lateral commands (1 = off, lower = smoother)
`arc_visualization_length`	`2.0`	m, RViz curvature arc length
`marker_lifetime_sec`	`0.5`	s, RViz marker lifetime
`avoidance_enabled`	`true`	blend obstacle avoidance into cmd_vel

Parameter	Default	Meaning
`avoidance_timeout_sec`	`0.5`	s, stop if navigation/avoidance goes silent (fail-safe)
`tf_timeout_sec`	`0.5`	s, stop if map->base_link stops updating (odometry died)
`estop_release_sec`	`1.0`	s, path must stay clear this long before resuming after an e-stop
`blocked_timeout_sec`	`30.0`	s, report BLOCKED when no progress is possible for this long
`off_path_slow_distance`	`0.6`	m, cross-track error that caps speed to min_linear_velocity
`off_path_stop_distance`	`1.5`	m, cross-track error that stops until navigation is restarted
`loop_route`	`true`	patrol: restart the route when it ends (closed routes only)
`loop_close_distance`	`1.0`	m, first/last waypoint gap that still counts as a closed loop
`low_battery_percentage`	`0.2`	0-1, below this the patrol stops at the end of the current lap
`stop_point_skip_margin`	`5`	waypoints, progress past a stop point before it is skipped
`holonomic`	`true`	true: close x/y/yaw together (legged/omni), false: turn-drive-turn
`avoidance_lateral_enabled`	`true`	holonomic only: apply the avoidance sideways push to linear.y
`approach_radius`	`0.5`	m, distance to a stop point that switches to the final approach
`approach_timeout_sec`	`20.0`	s, give up converging and dwell where the robot is
`join_max_distance`	`3.0`	m, a robot started off the route walks onto it from here (0: off)
`join_timeout_sec`	`60.0`	s, give up walking onto the route and follow from where it is

Parameter	Default	Meaning
`approach.kp_xy`	`1.2`	P gain on position error
`approach.kp_yaw`	`1.5`	P gain on yaw error
`approach.max_speed`	`0.2`	m/s
`approach.max_yaw_rate`	`0.6`	rad/s
`approach.min_speed`	`0.04`	m/s, floor to beat the robot's joystick deadband
`approach.position_tolerance`	`0.05`	m
`approach.yaw_tolerance_deg`	`3.0`	deg
`auto_mode_on_service`	`""`	std_srvs/Trigger services called on navigation start/stop ("" = skip). e.g. robot driver services that hand motion control to the SDK and back to the vendor remote
`auto_mode_off_service`	`""`	std_srvs/Trigger service called when navigation stops ("" = skip)
`inspection_services`	`["/anomaly_detector_node/check_now"]`	std_srvs/Trigger services called at every stop point after aligning ("" = none), e.g. ["/anomaly_detector_node/check_now"]. The robot waits for them up to inspection_timeout_sec.
`inspection_timeout_sec`	`20.0`	VLM inference on the Jetson takes a few seconds
`require_localization`	`true`	refuse start_nav until localization/status reports a source
`controller`	`holonomic`	holonomic (legged/omni robots: walks forward, sideways or backwards, heading kept along the route) | pure_pursuit | pid | mppi | lqr | stanley (steering only: forward + yaw)
`holonomic.lookahead_distance`	`0.6`	m, carrot ahead along the route; longer = gentler rejoin
`holonomic.heading_lookahead`	`0.6`	m, the heading follows the route direction this far ahead
`holonomic.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance

Parameter	Default	Meaning
`holonomic.kp_yaw`	`1.5`	rad/s per rad of heading error
`holonomic.max_lateral_velocity`	`0.25`	m/s sideways (0 when holonomic: false)
`holonomic.max_reverse_velocity`	`0.15`	m/s backwards, when the route is behind the robot
`holonomic.min_alignment_scale`	`0.25`	translation speed factor while the heading is far off
`avoidance_mode`	`path_shift`	Obstacles, with the holonomic controller: path_shift: slide sideways around them along a line parallel to the route, heading kept, never more than path_shift.max_offset away; stop short when there is no gap; reactive: blend the obstacle_avoidance_node steering (the only mode of other controllers)
`footprint.front`	`0.40`	m, base_link to the front edge (also used by the control mux)
`footprint.rear`	`0.40`	m, base_link to the rear edge
`footprint.half_width`	`0.28`	m, base_link to the side edges
`path_shift.max_offset`	`0.8`	m, furthest the robot leaves the route to pass something
`path_shift.look_ahead`	`2.0`	m of route ahead of the front edge that must be free
`path_shift.clearance`	`0.10`	m, gap kept between the robot's side and an obstacle
`path_shift.offset_rate`	`0.3`	m/s, how fast it slides onto a new offset
`path_shift.switch_cost`	`0.5`	hysteresis: higher = keeps its side longer
`path_shift.switch_margin`	`0.05`	m, extra clearance a new offset needs (no flicker)
`path_shift.return_delay_sec`	`0.5`	s, a way back toward the route must stay free this long
`path_shift.decel`	`0.5`	m/s^2, braking used to slow down toward an obstacle

Parameter	Default	Meaning
`path_shift.stop_margin`	`0.40`	m, stop this far before an obstacle (front edge)
`pure_pursuit.lookahead_distance`	`1.0`	m, look-ahead point on the route
`pure_pursuit.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance
`pid.lookahead_distance`	`1.0`	m, look-ahead point on the route
`pid.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance
`pid.kp`	`1.5`	proportional gain on the heading error
`pid.ki`	`0.0`	integral gain on the heading error
`pid.kd`	`0.2`	derivative gain on the heading error
`mppi.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance
`mppi.horizon_steps`	`15`	steps per rollout
`mppi.dt`	`0.1`	s per rollout step
`mppi.num_samples`	`200`	rollouts per tick
`mppi.angular_std`	`1.0`	rad/s, sampling noise
`mppi.temperature`	`0.05`	softmax sharpness for the normalized cost
`mppi.window_points`	`60`	path points ahead used for the tracking cost
`lqr.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance
`lqr.q_lateral`	`1.0`	weight on cross-track error
`lqr.q_heading`	`0.5`	weight on heading error
`lqr.r_angular`	`0.5`	weight on angular effort (higher = gentler)
`stanley.goal_tolerance`	`0.3`	m, the end of an open route counts as reached within this distance
`stanley.k_cross_track`	`1.0`	cross-track gain
`stanley.k_soft`	`0.2`	m/s, softening so low speeds do not over-steer

Parameter	Default	Meaning
`stanley.k_heading`	`1.5`	steering angle to yaw-rate gain
`speed_regulator_enabled`	`false`	slow down on sharp turns
`speed_regulator.kp`	`0.3`	proportional gain of the turn slow-down
`speed_regulator.ki`	`0.0`	integral gain of the turn slow-down
`speed_regulator.kd`	`0.0`	derivative gain of the turn slow-down
`speed_regulator.min_scale`	`0.3`	lowest speed factor in a sharp turn

`obstacle_avoidance_node`

Parameter	Default	Meaning
`algorithm`	`vfh`	braitenberg | vfh | potential_field | follow_gap
`safe_distance`	`1.0`	m from base_link, obstacles closer than this shape the command
`emergency_distance`	`0.75`	m from base_link (~0.4 m from the robot's nose), hard stop
`emergency_cone_deg`	`60.0`	deg, frontal cone checked for hard stop / blocked front
`braitenberg.steering_gain`	`1.5`	obstacle-to-steering strength
`braitenberg.velocity_gain`	`1.0`	obstacle-to-slow-down strength
`braitenberg.steering_zone_center_deg`	`45.0`	deg, centre of the left and right zones that steer
`braitenberg.steering_zone_width_deg`	`60.0`	deg, width of those steering zones
`braitenberg.velocity_zone_center_deg`	`60.0`	deg, centre of the left and right zones that slow down
`braitenberg.velocity_zone_width_deg`	`90.0`	deg, width of those slow-down zones
`braitenberg.following_zone_width_deg`	`30.0`	deg, frontal zone where obstacles slow the robot
`braitenberg.smoothing`	`0.85`	0-1 output filter (higher = smoother/slower)

Parameter	Default	Meaning
`vfh.min_gap_width_deg`	`20.0`	deg, narrowest passable gap
`vfh.steering_gain`	`1.0`	gap-heading-to-steering strength
`vfh.velocity_gain`	`1.0`	frontal-obstacle-to-speed strength
`vfh.smoothing`	`0.85`	0–1 output filter (higher = smoother/slower)
`potential_field.repulsion_gain`	`0.5`	how hard obstacles push the robot away
`potential_field.steering_gain`	`0.8`	obstacle-to-steering strength
`potential_field.lateral_gain`	`0.3`	repulsion to sideways m/s (holonomic robots)
`potential_field.max_lateral`	`0.2`	m/s cap on the sideways push
`potential_field.velocity_gain`	`1.0`	obstacle-to-slow-down strength
`potential_field.smoothing`	`0.8`	0–1 output filter (higher = smoother/slower)
`follow_gap.bubble_radius`	`0.45`	m, safety bubble around the nearest obstacle
`follow_gap.steering_gain`	`1.0`	obstacle-to-steering strength
`follow_gap.velocity_gain`	`1.0`	obstacle-to-slow-down strength
`follow_gap.smoothing`	`0.85`	0–1 output filter (higher = smoother/slower)

`path_loader_node`

Parameter	Default	Meaning
`waypoints_file`	`""`	route CSV (x,y,yaw_deg,dwell_sec)
`route_state_file`	`~/.pashupati/last_route`	last route loaded via navigation/load_path, reloaded at start when waypoints_file is empty
`republish_period`	`2.0`	s, RViz-only Path republish (0 = off); waypoints are latched
`smoothing_enabled`	`true`	resample + smooth between stop points (stop points never move)
`resample_spacing`	`0.1`	m, uniform point spacing after resampling

Parameter	Default	Meaning
`smooth_weight_data`	`0.05`	pull toward the recording (higher = closer, noisier)
`smooth_weight_smooth`	`0.4`	pull toward neighbours (higher = smoother), keep < 0.5
`max_smoothing_deviation`	`0.3`	m, warn when smoothing moves the route further than this

16.2 robot_bridge

control mux, protective field, handheld, video · `src/robot_bridge/config/bridge_params.yaml`

`control_mux_node`

Parameter	Default	Meaning
`mode`	`auto`	auto (navigation drives) | remote (teleop drives)
`remote_timeout_sec`	`0.3`	zero the robot when teleop goes silent
`nav_timeout_sec`	`0.5`	zero the robot when navigation goes silent in auto mode
`pause_nav_on_remote`	`true`	pause navigation while the remote drives
`estop_hold_hz`	`10.0`	zero commands per second while the e-stop is engaged
`max_vx`	`1.0`	m/s
`max_vy`	`0.5`	m/s
`max_wz`	`1.5`	rad/s
`field_in_auto`	`true`	Protective field: limit every command to a speed that still stops `margin` before the nearest obstacle in the swept path (perception/ safety_scan, 360 deg).
`field_in_remote`	`false`	REMOTE: the operator watches the robot, nothing limits it
`safety_scan_timeout_sec`	`0.5`	older scan: auto mode stops, remote mode creeps

Parameter	Default	Meaning
`stale_remote_speed`	`0.15`	m/s cap for the handheld while the scan is stale
`field.front`	`0.40`	m, base_link to the front edge
`field.rear`	`0.40`	m, base_link to the rear edge
`field.half_width`	`0.28`	m, base_link to the side edges
`field.max_decel`	`0.6`	m/s^2, braking the robot reliably achieves
`field.reaction_time`	`0.25`	s, sensing + command latency
`field.margin`	`0.15`	m, gap kept to the obstacle
`field.lateral_margin`	`0.10`	m, extra width of the swept path
`field.min_speed`	`0.02`	m/s, slower commands are not limited

`teleop_udp_node`

Parameter	Default	Meaning
`port`	`9870`	UDP port of the handheld link and of robot discovery
`token`	`""`	shared secret; when set every message must carry "k"
`robot_name`	`""`	name in discovery beacons (default: hostname)
`client_timeout_sec`	`2.0`	s without messages before the handheld counts as gone (its commands stop)
`max_vx`	`1.0`	m/s, clamp on teleop commands
`max_vy`	`0.5`	m/s, clamp on sideways teleop commands
`max_wz`	`1.5`	rad/s, clamp on teleop turning
`telemetry_hz`	`10.0`	state messages per second to the handheld
`robot_command_prefix`	`/robot/`	driver services; robot.launch.py sets /<robot_ns>/

Parameter	Default	Meaning
`robot_commands`	`[stand_up, sit_down, emergency_stop]`	Commands shown on the handheld, each a std_srvs/Trigger <prefix><name> of the robot driver. Add the robot's own (gait, speed modes, …) in the site file, see src/xtras/README.md.
`routes_dir`	`/home/robot/dev/Pashupati/map`	where the handheld lists and loads routes from
`map_frame`	`map`	TF frame name
`base_frame`	`base_link`	TF frame name
`camera_topic`	`/camera/camera/color/image_raw/compressed`	camera stream for the JPEG video
`camera_compressed`	`true`	false: raw rgb8/bgr8 Image, JPEG-encoded here
`camera_fps`	`8.0`	JPEG frames per second (when there is no WebRTC)
`camera_jpeg_quality`	`60`	JPEG quality of re-encoded frames (0-100)
`camera_max_width`	`640`	px, wider frames are scaled down
`camera_max_bytes`	`60000`	bigger driver JPEGs are re-encoded (needs OpenCV)
`video_port`	`0`	WebRTC video (MediaMTX WHEP port, robot.launch.py sets it when MediaMTX runs); 0: the handheld only gets the JPEG frames above
`video_path`	`camera`	MediaMTX path of the WebRTC video
`tags_topic`	`/perception/apriltag/detections`	AprilTags drawn over the handheld video (JPEG or WebRTC); "" turns it off
`tags_camera_info_topic`	`/perception/apriltag/camera_info`	camera model for the AprilTag outlines

Parameter	Default	Meaning
`lidar_topic`	`/cloud_registered_body`	FAST-LIO body-frame cloud
`lidar_hz`	`2.0`	LiDAR frames per second for the handheld map
`lidar_voxel`	`0.1`	m, LiDAR points are thinned to one per cube this size
`lidar_max_points`	`3000`	most LiDAR points per frame to the handheld
`inspection_node`	`/anomaly_detector_node`	enabled, model, interval_sec; status topic
`people_node`	`/person_detector_node`	enabled; perception/people_status
`navigation_node`	`/path_follower_node`	target_linear_velocity up to max_linear_velocity
`patrol_speed_min`	`0.1`	m/s, slowest patrol speed the handheld may set

`video_stream_node`

Parameter	Default	Meaning
`image_topic`	`/camera/camera/color/image_raw`	raw rgb8/bgr8/mono8/yuv422 image
`fps`	`15.0`	video frames per second
`max_width`	`1280`	wider camera images are scaled down
`bitrate_kbps`	`1500`	H.264 bitrate; lower it on weak Wi-Fi
`keyframe_sec`	`1.0`	a viewer starts at a key frame: keep it short
`encoder`	`auto`	auto: the first that works of nvv4l2h264enc (Jetson hardware), x264enc, openh264enc

16.3 robot_perception

LiDAR sectors, people, AprilTags · `src/robot_perception/config/perception_params.yaml`

lidar_sector_node

Parameter	Default	Meaning
`input_type`	`livox`	livox: Livox CustomMsg on livox/lidar, read straight from the serialized bytes (~1 ms per scan, works without FAST-LIO). Use this on a Livox robot. pointcloud2: any PointCloud2, e.g. FAST-LIO /cloud_registered_body. That cloud has FAST-LIO's `blind` radius cut out, so obstacles right next to the robot are missed.
`pointcloud_topic`	`/cloud_registered_body`	cloud used with input_type pointcloud2
`base_frame`	`base_link`	TF frame name
`lidar_frame`	`livox_frame`	used when the cloud header has no frame_id
`num_sectors`	`32`	number of angular sectors used to divide the FOV
`fov_deg`	`180.0`	total horizontal field of view in degrees
`range_max`	`8.0`	maximum LiDAR detection range in meters
`range_min`	`0.2`	minimum LiDAR detection range in meters
`obstacle_z_min`	`-0.25`	Obstacle height filtering, relative to base_link (body center, NOT the floor). base_link is ~0.35 m above the floor when standing: z_min -0.25 keeps anything taller than ~10 cm (bags, kids, low planters) while dropping floor returns. Measure and tune.
`obstacle_z_max`	`1.20`	m above base_link; higher points are ignored
`self_filter_x_min`	`-0.40`	Robot footprint in base_link; points inside are the robot's own legs/body and are ignored
`self_filter_x_max`	`0.40`	m, front of the robot body box
`self_filter_y_min`	`-0.28`	m, right side of the robot body box

Parameter	Default	Meaning
`self_filter_y_max`	`0.28`	m, left side of the robot body box
`safety_scan_enabled`	`true`	360 deg safety scan for the protective field in control_mux_node (perception/safety_scan)
`safety_num_sectors`	`72`	5 deg sectors
`safety_range_max`	`4.0`	m, beyond this nothing limits the speed
`min_raw_points`	`100`	fewer points per cloud: LiDAR covered/failed, report all blocked
`publish_markers`	`true`	publish sector rays and detected obstacle points
`marker_lifetime_sec`	`0.5`	lifetime of each marker in seconds
`corridor_x_min`	`0.0`	minimum X boundary in base_link frame in meters
`corridor_x_max`	`5.0`	maximum X boundary in base_link frame in meters
`corridor_y_min`	`-1.0`	minimum Y boundary in base_link frame in meters
`corridor_y_max`	`1.0`	maximum Y boundary in base_link frame in meters

`person_detector_node`

Parameter	Default	Meaning
`enabled`	`true`	live (handheld, ros2 param set); PEOPLE=false: not started at all
`state_file`	`~/.pashupati/people.json`	live on/off kept across restarts; "": off
`image_topic`	`/camera/camera/color/image_raw`	raw: no JPEG encoding in the camera driver
`compressed`	`false`	true: image_topic carries JPEG (CompressedImage)

Parameter	Default	Meaning
`camera_info_topic`	`/camera/camera/color/camera_info`	camera model, to place people on the floor
`model_path`	`/home/robot/dev/Pashupati/models/yolo11n-416.onnx`	YOLO11n (the smallest YOLO) exported to ONNX at 416 px: script/setup.sh install people_model. .onnx runs without PyTorch (ONNX Runtime, CUDA when that build is installed, else OpenCV); .pt / .engine go through Ultralytics.
`backend`	`auto`	auto | onnxruntime | opencv | ultralytics
`threads`	`2`	CPU threads for one inference
`confidence`	`0.4`	live
`image_size`	`416`	OpenCV / Ultralytics input size (ONNX Runtime reads the model's)
`rate_hz`	`0.0`	live: detections per second; 0 = 5 on the GPU, 2 on the CPU
`floor_z`	`-0.35`	floor height in base_link (measure: -height of base_link)
`max_range`	`12.0`	m, ignore people further away (feet projection gets noisy)
`map_frame`	`map`	TF frame name
`base_frame`	`base_link`	TF frame name

apriltag_throttle

Parameter	Default	Meaning
`rate_hz`	`5.0`	camera frames per second given to the AprilTag detector

apriltag

Parameter	Default	Meaning
`image_transport`	`raw`	image transport type: "raw" or "compressed"

Parameter	Default	Meaning
`family`	`36h11`	AprilTag family: 16h5, 25h9, 36h11, etc.
`size`	`0.16`	default physical tag edge length in meters
`profile`	`false`	print detector profiling information
`max_hamming`	`0`	maximum allowed bit errors in the detected tag
`detector.threads`	`1`	number of CPU threads used by the detector
`detector.decimate`	`2.0`	image downsampling factor for faster detection
`detector.blur`	`0.0`	Gaussian blur sigma applied during detection
`detector.refine`	`true`	refine detected tag corner positions
`detector.sharpening`	`0.25`	sharpening strength applied before decoding
`detector.debug`	`false`	save additional detector debug images
`pose_estimation_method`	`pnp`	pose estimation method: PnP
`tag.ids`	`[2, 4]`	AprilTag IDs whose TF frames should be published
`tag.frames`	`[base_map, charging_dock]`	TF frame assigned to each tag ID
`tag.sizes`	`[0.15, 0.15]`	physical tag size for each tag ID in meters

16.4 robot_localization

map frame from AprilTags · `src/robot_localization/config/localization_params.yaml`

`localization_node`

Parameter	Default	Meaning
`tags_config_file`	`""`	JSON {tag_frame: {x,y,z,qx,qy,qz,qw}} in map frame
`map_frame`	`map`	TF frame name
`odom_frame`	`odom`	TF frame name
`base_frame`	`base_link`	TF frame name

Parameter	Default	Meaning
`lio_odom_frame`	`camera_init`	FAST-LIO world frame
`lio_body_frame`	`body`	FAST-LIO body frame (= Livox IMU)
`imu_frame`	`livox_imu_frame`	same physical frame as `body`, from the URDF
`camera_frame`	`camera_color_optical_frame`	TF frame of the camera the tags are seen in
`tag_max_age_sec`	`0.5`	ignore tag TFs older than this
`auto_localize_on_start`	`true`	retry tag localization every 1 s until it succeeds
`assume_start_origin`	`false`	true: the start pose is the map origin (no tag needed)
`tag_frames`	`[base_map, charging_dock]`	tag TF frames from perception_params.yaml tag.frames
`tag_record_duration_sec`	`2.0`	s of averaging per tag recording
`tags_record_file`	`""`	empty: overwrite tags_config_file (a .bak is kept)
`auto_correct_period_sec`	`0.0`	>0: re-correct drift from any visible tag every N s
`max_correction_m`	`0.5`	reject auto corrections larger than this
`max_correction_deg`	`15.0`	deg, reject auto corrections that turn more
`map_odom_hz`	`20.0`	map->odom is republished on /tf at this rate

16.5 robot_inspection

AI inspection · `src/robot_inspection/config/inspection_params.yaml`

`anomaly_detector_node`

Parameter	Default	Meaning
`enabled`	`true`	live: false stops inspecting (stop points answer "inspection off")

Parameter	Default	Meaning
`state_file`	`~/.pashupati/inspection.json`	live changes (handheld) kept across restarts; "": off
`image_topic`	`/camera/camera/color/image_raw`	sensor_msgs/Image (rgb8, bgr8, rgba8, bgra8, mono8)
`ollama_url`	`http://127.0.0.1:11434`	address of the Ollama server
`model`	`qwen3-vl:2b-instruct`	live: any Ollama vision model, see models
`models`	`[qwen3-vl:2b-instruct, moondream:latest, ahmadwaqar/smolvlm2-2.2b-instruct:latest]`	offered on the handheld (plus every other vision model in Ollama)
`categories`	`[trash, spill, stain]`	what the model chooses from (plus "none"), a-z 0-9 _
`alarm_categories`	`[trash, spill]`	only these are reported; stain is logged but not reported
`category_descriptions`	`[""]`	own text per type, "name: what to look for" (English works best); trash, spill, stain, floor_damage and fallen_person have built-in texts
`use_schema`	`true`	false: free-text answer (FLOOR:/TYPE:) parsed manually
`describe`	`false`	true: model also writes one sentence about the floor
`debug`	`false`	true: log every inspection and the raw model answer
`keep_alive`	`"-1"`	keep the model in memory: "-1" forever, or e.g. "30m" (quoted)
`num_ctx`	`2048`	tokens of context the model is loaded with: the image and the type list must fit
`timeout_sec`	`30.0`	s, longest wait for one answer of the model

Parameter	Default	Meaning
`max_image_age_sec`	`2.0`	s, older camera frames are not inspected
`max_image_width`	`640`	downscale before inference
`interval_sec`	`0.0`	live: 0 back to back, > 0 every N s, < 0 only at stop points
`cooldown_sec`	`60.0`	periodic mode: do not report a category again within this time
`queue_dir`	`~/.pashupati/inspection_queue`	kept across restarts; "": in memory only
`queue_max`	`50`	full: the oldest frame is dropped for the new one
`queue_min_change`	`12.0`	grey levels part of the picture must change to queue it again (standing still adds nothing); 0: every frame

16.6 robot_odometry

odometry start, monitor, route recording · `src/robot_odometry/config/odometry_params.yaml`

odometry_gate

Parameter	Default	Meaning
`imu_topic`	`/livox/imu`	IMU of the LiDAR, watched for stillness
`still_sec`	`2.0`	this long without motion before FAST-LIO may start
`max_gyro_rad_s`	`0.03`	rad/s, more rotation counts as moving
`max_acc_rel_std`	`0.01`	std of |acc| / mean |acc|, unit-free (Livox reports g)
`max_tilt_deg`	`8.0`	gravity vs Mid-360 z; catches a sitting robot. 0: off
`timeout_sec`	`60.0`	start anyway after this long (with an error). 0: wait forever

odometry_monitor_node

Parameter	Default	Meaning
odom_topic	/Odometry	LiDAR odometry to watch
leg_odom_topic	/robot/odom	robot's own speed for the cross-check, "" to disable
stale_sec	0.5	FAST-LIO runs at 10 Hz
max_speed_mps	3.0	faster than this between two frames is a jump
max_z_drift_m	0.5	flat floor: big z means a tilted start. 0: off
max_speed_diff_mps	0.35	m/s, LiDAR and leg odometry speeds further apart are a mismatch
speed_diff_sec	1.5	mismatch must last this long (degenerate corridor, slip)
min_check_speed_mps	0.1	m/s, slower: speeds are not compared
restart_origin_radius_m	0.05	m, odometry jumping back this close to its origin counts as a restart
restart_min_offset_m	0.3	m, ... when the robot was at least this far from the origin before

route_recorder_node

Parameter	Default	Meaning
output_path	"/home/robot/dev/Pashupati/map/"	writes map/<date>/<time>_path.csv
map_frame	"map"	TF frame name
base_frame	"base_link"	TF frame name
min_waypoint_spacing_m	0.2	m, a new route point only after moving this far
record_rate_hz	10.0	pose samples per second while recording

16.7 robot_health

health monitor · `src/robot_health/config/health_params.yaml`

health_monitor_node

Parameter	Default	Meaning
`robot_ns`	`robot`	fills {robot_ns} in topics (robot.launch.py sets it)
`topics`	`[/livox/lidar:5.0, /Odometry:8.0, /perception/sector_scan:5.0, /camera/camera/color/camera_info:5.0, /{robot_ns}/odom:10.0]`	"topic:minimum Hz"; below 80 % of it is an issue
`max_cpu_temp_c`	`85.0`	deg C, hotter is an issue
`max_memory_percent`	`92.0`	%, more memory in use is an issue
`min_disk_free_gb`	`5.0`	GB, less free disk is an issue
`disk_path`	`"~"`	the disk that is checked
`publish_hz`	`1.0`	health/status messages per second
`expected_nodes`	`[""]`	node names that must be running (robot.launch.py fills it)
`startup_grace_sec`	`30.0`	no missing-node issues during start-up

16.8 robot_fleet

NETRA uplink · `src/robot_fleet/config/fleet_params.yaml`

`fleet_uplink_node`

Parameter	Default	Meaning
`server_url`	`""`	e.g. https://netra.example.com (empty: spool only)
`token`	`""`	robot token issued by the command center
`robot_id`	`robot-01`	robot name in NETRA (ROBOT_ID of the env file wins)
`verify_tls`	`true`	check the server certificate
`telemetry_hz`	`1.0`	telemetry messages per second to NETRA
`upload_period_sec`	`5.0`	s between upload attempts of waiting files
`spool_dir`	`~/.pashupati/spool`	anomalies and routes wait here until the server accepts them
`routes_dir`	`/home/robot/dev/Pashupati/map`	recorded routes, uploaded to NETRA
`map_frame`	`map`	TF frame name
`base_frame`	`base_link`	TF frame name
`crowd_cell_size`	`1.0`	m, heatmap grid in the map frame
`crowd_window_sec`	`60.0`	one crowd sample per window (peak people per cell)
`anomaly_topic`	`/anomaly_detector/result`	inspection results to report
`anomaly_image_topic`	`/camera/camera/color/image_raw/compressed`	fallback frame

Parameter	Default	Meaning
`anomaly_evidence_topic`	`/anomaly_detector/evidence`	the frame each result was detected in
`anomaly_min_confidence`	`0.5`	0-1, less confident results are dropped